

Consider the following for the **two (02)** items that follow :

The probabilities that A , B and C become managers are $3/10$, $1/2$ and $4/5$ respectively. The probabilities that bonus scheme will be introduced if A , B and C become managers are $4/9$, $2/9$ and $1/3$ respectively.

111. What is the probability that the bonus scheme will be introduced?

- (a) $17/45$ (b) $19/45$
(c) $23/45$ (d) $26/45$

112. If the bonus scheme has been introduced, then what is the probability that the manager appointed was B ?

- (a) $5/23$ (b) $6/23$
(c) $7/23$ (d) $8/23$

113. The arithmetic mean of 100 observations is 50. If 5 is subtracted from each observation and then divided by 20, then what is the new arithmetic mean?

- (a) 2.25 (b) 3.5
(c) 4.25 (d) 5.5

114. The standard deviation of 100 observations is 10. If 5 is added to each observation and then divided by 20, then what will be the new standard deviation?

- (a) 0.25 (b) 0.5
(c) 0.75 (d) 1.00

115. If $P(A) = 1/3$, $P(B) = 1/2$ and $P(A \cap B) = 1/4$, then what is the value of $P(B|A^c)$?

- (a) $1/8$ (b) $3/8$
(c) $5/8$ (d) $7/8$

116. If $P(A) = 1/3$, $P(B) = 1/2$ and $P(A \cap B) = 1/4$, then what is the value of $P(A^c \cap B^c)$?

- (a) $1/4$ (b) $5/12$
(c) $7/12$ (d) $11/12$

117. If two fair dice are tossed, then what is the probability that the sum of the numbers on the faces of the dice is strictly greater than 7?

- (a) $1/3$ (b) $5/12$
(c) $7/12$ (d) $3/4$

118. The probability of a man hitting a target is $1/5$. If the man fires 7 times, then what is the probability that he hits the target at least twice?

- (a) $1 - \left(\frac{3}{5}\right)\left(\frac{4}{5}\right)^6$
(b) $1 - \left(\frac{3}{5}\right)\left(\frac{4}{5}\right)^7$
(c) $1 - \left(\frac{11}{5}\right)\left(\frac{4}{5}\right)^6$
(d) $1 - \left(\frac{11}{5}\right)\left(\frac{4}{5}\right)^7$

119. Let X be a random variable following binomial distribution whose mean and variance are 200 and 160 respectively. What is the value of the number of trials (n)?

- (a) 500 (b) 1000
(c) 1500 (d) 2000

120. What is the arithmetic mean of $8^2, 9^2, 10^2, \dots, 15^2$?

- (a) 133.5 (b) 135.5
(c) 137.5 (d) 139.5

SPACE FOR ROUGH WORK